

Supplementary Material

Demographic Disparities in Criminal Sentencing: Evidence from Mongolian Courts

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Table A1. Holm-Bonferroni correction for Model 1

Hypothesis	Predictor	p	Holm threshold	Decision
H4 (Employment)	Employed	< .001	0.0125	Reject
H1 (Gender)	Female	< .001	0.0167	Reject
H3 (Age non-linear)	Age squared	.007	0.0250	Reject
H2 (Education)	Education level	.083	0.0500	Do not reject

Table A2. Variance decomposition

Block	R-squared	Delta R-squared
Demographics only	0.039	0.039
+ Crime + prior criminal	0.072	0.033
+ Year/Court FE (= Model 1)	0.080	0.008

Table A3. Two-part imprisonment model, stage 1

Predictor	Logit B	SE	z	p	Odds ratio	95% OR CI
Female	-0.538	0.054	-9.90	< .001	0.58	0.52 to 0.65
Education level	-0.147	0.016	-9.37	< .001	0.86	0.84 to 0.89
Employed	-0.923	0.034	-26.97	< .001	0.40	0.37 to 0.42
Prior criminal record	1.051	0.034	30.66	< .001	2.86	2.67 to 3.06

Table A4. Full robustness estimates

Specification	Predictor	B	SE	p	95% CI	R-squared	N	Note
Sample B (no age)	Female	-1.483	0.250	< .001	-1.974 to -0.992	0.078	35,995	Model 1 without age/age_sq
Sample B (no age)	Education level	-0.146	0.108	.178	-0.357 to 0.066	0.078	35,995	Model 1 without age/age_sq
Sample B (no age)	Employed	-6.101	0.652	< .001	-7.380 to -4.822	0.078	35,995	Model 1 without age/age_sq
Sample B (no age)	Prior criminal record	5.549	0.425	< .001	4.716 to 6.382	0.078	35,995	Model 1 without age/age_sq
2020-2022	Female	-2.029	0.463	< .001	-2.936 to -1.123	0.082	15,976	Model 1 temporal subset
2020-2022	Education level	-0.177	0.212	.406	-0.593 to 0.240	0.082	15,976	Model 1 temporal subset
2020-2022	Employed	-6.837	0.655	< .001	-8.120 to -5.554	0.082	15,976	Model 1 temporal subset
2020-2022	Prior criminal record	5.880	0.561	< .001	4.780 to 6.981	0.082	15,976	Model 1 temporal subset
2023-2025	Female	-1.038	0.606	.087	-2.225 to 0.149	0.083	13,069	Model 1 temporal subset
2023-2025	Education level	-0.269	0.146	.065	-0.555 to 0.016	0.083	13,069	Model 1 temporal subset
2023-2025	Employed	-5.477	0.864	< .001	-7.170 to -3.785	0.083	13,069	Model 1 temporal subset
2023-2025	Prior criminal record	4.702	0.474	< .001	3.774 to 5.630	0.083	13,069	Model 1 temporal subset
Log-severity	Female	-0.208	0.025	< .001	-0.257 to -0.159	0.213	28,959	DV = log(severity); coefficients are log units
Log-severity	Education level	0.022	0.010	.024	0.003 to 0.041	0.213	28,959	DV = log(severity); coefficients are log units
Log-severity	Employed	-0.385	0.040	< .001	-0.463 to -0.306	0.213	28,959	DV = log(severity); coefficients are log units
Log-severity	Prior criminal record	0.496	0.027	< .001	0.443 to 0.550	0.213	28,959	DV = log(severity); coefficients are log units
Two-part stage 2	Female	3.784	1.795	.035	0.266 to 7.302	0.210	5,521	OLS on imprisoned cases with sentence_months
Two-part stage 2	Education level	-0.604	0.445	.175	-1.477 to 0.268	0.210	5,521	OLS on imprisoned cases with sentence_months
Two-part stage 2	Employed	-4.306	1.019	< .001	-6.302 to -2.309	0.210	5,521	OLS on imprisoned cases with sentence_months
Two-part stage 2	Prior criminal record	0.381	1.223	.755	-2.016 to 2.778	0.210	5,521	OLS on imprisoned cases with sentence_months
Legal controls subsample	Female	-0.044	0.777	.955	-1.567 to 1.478	0.343	6,334	Replaces agg/mit with victim_minor, injury_serious, intoxicated, restitution; non-representative subsample
Legal controls subsample	Education level	-0.736	0.303	.015	-1.329 to -0.143	0.343	6,334	Replaces agg/mit with victim_minor, injury_serious, intoxicated, restitution; non-representative subsample
Legal controls subsample	Employed	-8.165	0.851	< .001	-9.833 to -6.496	0.343	6,334	Replaces agg/mit with victim_minor, injury_serious, intoxicated, restitution; non-representative subsample
Legal controls subsample	Prior criminal record	4.154	0.627	< .001	2.926 to 5.382	0.343	6,334	Replaces agg/mit with victim_minor, injury_serious, intoxicated, restitution; non-representative subsample
Severity: Article 5.4 community conversion	Female	-1.492	0.291	< .001	-2.063 to -0.921	0.083	29,132	Community service = hours/240; other default weights unchanged
Severity: Article 5.4 community conversion	Education level	-0.199	0.134	.139	-0.463 to 0.064	0.083	29,132	Community service = hours/240; other default weights unchanged
Severity: Article 5.4 community conversion	Employed	-6.109	0.611	< .001	-7.307 to -4.910	0.083	29,132	Community service = hours/240; other default weights unchanged
Severity: Article 5.4 community conversion	Prior criminal record	5.242	0.431	< .001	4.398 to 6.086	0.083	29,132	Community service = hours/240; other default weights unchanged
Severity: higher noncustodial weights	Female	-1.436	0.310	< .001	-2.044 to -0.829	0.080	29,132	Fine divisor 300,000; suspended weight 0.7; community service hours/160
Severity: higher noncustodial weights	Education level	0.098	0.139	.482	-0.175 to 0.370	0.080	29,132	Fine divisor 300,000; suspended weight 0.7; community service hours/160
Severity: higher noncustodial weights	Employed	-5.740	0.608	< .001	-6.931 to -4.549	0.080	29,132	Fine divisor 300,000; suspended weight 0.7; community service hours/160
Severity: higher noncustodial weights	Prior criminal record	5.162	0.438	< .001	4.303 to 6.020	0.080	29,132	Fine divisor 300,000; suspended weight 0.7; community service hours/160
Severity: lower noncustodial weights	Female	-1.505	0.283	< .001	-2.061 to -0.950	0.086	29,132	Fine divisor 600,000; suspended weight 0.3; community service hours/160
Severity: lower noncustodial weights	Education level	-0.355	0.133	.008	-0.616 to -0.094	0.086	29,132	Fine divisor 600,000; suspended weight 0.3; community service hours/160
Severity: lower noncustodial weights	Employed	-6.337	0.618	< .001	-7.548 to -5.126	0.086	29,132	Fine divisor 600,000; suspended weight 0.3; community service hours/160
Severity: lower noncustodial weights	Prior criminal record	5.340	0.429	< .001	4.500 to 6.180	0.086	29,132	Fine divisor 600,000; suspended weight 0.3; community service hours/160
Preregistered HC3 estimator	Female	-1.511	0.310	< .001	-2.119 to -0.904	0.080	29,132	Model 1 with HC3 rather than court-clustered SE
Preregistered HC3 estimator	Education level	-0.237	0.113	.036	-0.458 to -0.015	0.080	29,132	Model 1 with HC3 rather than court-clustered SE
Preregistered HC3 estimator	Employed	-6.280	0.285	< .001	-6.839 to -5.721	0.080	29,132	Model 1 with HC3 rather than court-clustered SE
Preregistered HC3 estimator	Prior criminal record	5.318	0.262	< .001	4.805 to 5.831	0.080	29,132	Model 1 with HC3 rather than court-clustered SE
Fine-only log amount	Female	-0.069	0.022	.002	-0.112 to -0.026	0.269	20,880	DV = log(sentence_fine_mnt); within fine only
Fine-only log amount	Education level	0.106	0.010	< .001	0.086 to 0.126	0.269	20,880	DV = log(sentence_fine_mnt); within fine only
Fine-only log amount	Employed	0.034	0.016	.033	0.003 to 0.065	0.269	20,880	DV = log(sentence_fine_mnt); within fine only

Specification	Predictor	B	SE	p	95% CI	R-squared	N	Note
Fine-only log amount	Prior criminal record	0.170	0.012	< .001	0.148 to 0.193	0.269	20,880	DV = log(sentence_fine_mnt); within fine only
Imprisonment-only log months	Female	0.076	0.039	.051	-0.000 to 0.152	0.181	5,520	DV = log(sentence_months); within imprisonment only
Imprisonment-only log months	Education level	0.010	0.010	.335	-0.010 to 0.031	0.181	5,520	DV = log(sentence_months); within imprisonment only
Imprisonment-only log months	Employed	-0.099	0.021	< .001	-0.140 to -0.058	0.181	5,520	DV = log(sentence_months); within imprisonment only
Imprisonment-only log months	Prior criminal record	0.053	0.024	.028	0.006 to 0.101	0.181	5,520	DV = log(sentence_months); within imprisonment only
Exclude fines above personal statutory maximum	Female	-1.478	0.299	< .001	-2.065 to -0.892	0.081	29,124	Excludes fine cases above 40,000,000 MNT under Article 5.3.2-.3
Exclude fines above personal statutory maximum	Education level	-0.258	0.133	.053	-0.519 to 0.003	0.081	29,124	Excludes fine cases above 40,000,000 MNT under Article 5.3.2-.3
Exclude fines above personal statutory maximum	Employed	-6.234	0.624	< .001	-7.456 to -5.011	0.081	29,124	Excludes fine cases above 40,000,000 MNT under Article 5.3.2-.3
Exclude fines above personal statutory maximum	Prior criminal record	5.368	0.447	< .001	4.493 to 6.244	0.081	29,124	Excludes fine cases above 40,000,000 MNT under Article 5.3.2-.3
Fine-only log amount (statutory range)	Female	-0.065	0.022	.003	-0.108 to -0.022	0.272	20,872	DV = log(sentence_fine_mnt); excludes amounts above 40,000,000 MNT
Fine-only log amount (statutory range)	Education level	0.105	0.010	< .001	0.086 to 0.124	0.272	20,872	DV = log(sentence_fine_mnt); excludes amounts above 40,000,000 MNT
Fine-only log amount (statutory range)	Employed	0.041	0.016	.011	0.009 to 0.072	0.272	20,872	DV = log(sentence_fine_mnt); excludes amounts above 40,000,000 MNT
Fine-only log amount (statutory range)	Prior criminal record	0.174	0.012	< .001	0.150 to 0.197	0.272	20,872	DV = log(sentence_fine_mnt); excludes amounts above 40,000,000 MNT

Table A5. Sample registry

Sample	Description	N	Required fields
full	Full cleaned sample	69,527	None
sample_a	Primary confirmatory adult sample	29,132	female, age, education_level, employed, crime_category, prior_criminal, severity
sample_b	Robustness sample without age requirement; known juveniles excluded	35,995	female, education_level, employed, crime_category, prior_criminal, severity
two_part_stage1	Two-part imprisonment model stage 1 adult sample	33,600	female, age, education_level, employed, crime_category, prior_criminal
two_part_stage2	Two-part imprisonment model stage 2 sample	5,521	female, age, education_level, employed, crime_category, prior_criminal, sentence_months

Table A6. Validation metrics

Field	Metric	Value	N	Threshold	Gate result
Employment	prediction coverage on gold	0.958	191	—	Not applicable
Employment	accuracy	0.978	183	0.900	Pass
Employment	kappa	0.956	183	0.800	Pass
Sentence_type	accuracy	0.790	300	—	Not applicable
Sentence_type	kappa	0.750	300	—	Not applicable
Sentence_months	exact match rate	0.833	60	—	Not applicable
Sentence_months	within 1 month rate	0.867	60	—	Not applicable
Aggravating	present f1	0.150	283	0.850	Fail
Aggravating	count mae	1.035	283	1.000	Fail
Aggravating	count spearman r	0.203	283	—	Not applicable
Mitigating	present f1	0.665	283	0.850	Fail
Mitigating	count mae	1.933	283	1.000	Fail
Mitigating	count spearman r	0.409	283	—	Not applicable